

Problem Set 3

Given on: 26th November 2024
Due on: 9th December 2024
[No late submissions shall be accepted]
Total: 50 points + 5 bonus points
Submission Via: Hard Copy, no soft copies will be checked
Topics: Decidability, Reducibility & Time Complexity

Question 1 [10 Points]

Let B be the set of all infinite sequences over $\{0, 1\}$. Show that B is uncountable using a proof by diagonalization.

Question 2 [10 Points]

Let $L = \{ \langle G, s, t \rangle \mid G \text{ is a directed graph such that there is a path from } s \text{ to } t \}$. Prove that L is in class P .
Prove that L is in class NP .

Question 3 [10 Points]

Let $T = \{ a^i b^j c^k : i, j, k \in \mathbb{N} \}$. Prove that T is countable

Question 4 [10 Points]

If L_1 and L_2 are Turing recognizable languages, show that their concatenation is also Turing recognizable.
To do this, assume that you have Turing machines recognizing L_1 and L_2 . Then explain how to construct a Turing machine that recognizes $L_1 L_2$. Describe your construction in English. Feel free to give your Turing machine extra working tapes.

Question 5 [10 Points]

Let $T = \{ \langle M \rangle \mid M \text{ is a TM that accepts } w^{rev} \text{ whenever it accepts } w \}$. Show that T is undecidable.

Bonus Question [5 Points]

Let $T = \{ \langle G \rangle \mid G \text{ is a tree} \}$. Prove that T is in class P .